Companion
to

ChatGPT Teaches TikZ

Chapter 13: Decorations and Path Effects (I)

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13.1 What This Companion Adds

Chapter 13 introduced decorations that alter the appearance of a path without changing its underlying geometry. This Companion develops a controlled method for choosing a decoration, tuning its amplitude and segment length, and using braces as explanatory annotations.

Design

A decoration should encode meaning. If an ordinary line communicates the same information, the ordinary line is usually the clearer choice.

13.2 A Decoration Vocabulary

Option or decoration	Purpose
`decorate`	Activates a decoration on the path.
`coil`	Produces a spring-like path.
`zigzag`	Produces repeated sharp turns.
`snake`	Produces a smooth oscillating path.
`bumps`	Produces a repeated bumpy path.
`segment length`	Controls the repetition length.
`amplitude`	Controls the height of the effect.
`brace`	Replaces a path with a brace.
`mirror`	Places a brace on the opposite side.

Load both relevant libraries in the preamble:

```
\usetikzlibrary{
  decorations.pathmorphing,
  decorations.pathreplacing
}
```

13.3 The Anatomy of a Decorated Path

Every example begins with an ordinary path and adds two kinds of options.

```
\draw[
  decorate,
  decoration={coil}
] (0,0) -- (5,0);
```

The option `decorate` activates the mechanism. The `decoration` option selects and configures the effect.

Common Mistake

Writing `decoration={coil}` without `decorate` stores the choice but leaves the path ordinary.

13.4 Coils and Springs

A coil communicates flexibility, tension, or a spring-like connection.

```
\begin{tikzpicture}
  \draw[
    decorate,
    decoration={coil}
  ] (0,0) -- (5,0);
\end{tikzpicture}
```

See

The endpoints and direction are still those of the straight segment.

13.4.1 Tuning the Coil

```
\draw[
  decorate,
  decoration={
    coil,
    segment length=5mm,
    amplitude=2mm
  }
] (0,0) -- (5,0);
```

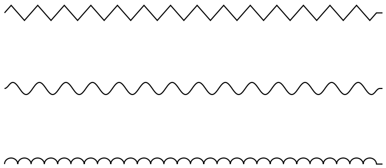
The segment length controls repetition frequency; amplitude controls height. Change one at a time when comparing effects.

13.5 Zigzags, Snakes, and Bumps

The same underlying segment can carry three distinct appearances.

```
\draw[decorate,decoration={zigzag}]
  (0,0) -- (5,0);
\draw[decorate,decoration={snake}]
  (0,-1) -- (5,-1);
\draw[decorate,decoration={bumps}]
  (0,-2) -- (5,-2);
```

See



Choose by role: a zigzag suggests repeated sharp change, a snake suggests smooth oscillation, and bumps suggest repeated local variation.

13.6 Comparing Parameters

Use the same endpoints while changing one option.

```
\draw[decorate,decoration={
  zigzag,segment length=3mm,amplitude=1mm
}] (0,0) -- (5,0);

\draw[decorate,decoration={
  zigzag,segment length=7mm,amplitude=1mm
}] (0,-1) -- (5,-1);
```

The equal amplitudes isolate the effect of segment length. For a fair visual comparison, avoid changing several parameters at once.

13.7 Braces as Annotations

Braces belong to the `decorations.pathreplacing` library. They group or measure existing objects.

```
\begin{tikzpicture}
  \node at (0,0) {$a$};
  \node at (1,0) {$b$};
  \node at (2,0) {$c$};

  \draw[
    decorate,
    decoration={brace,mirror,amplitude=5pt}
  ] (-.3,-.4) -- (2.3,-.4)
    node[midway,below=7pt] {three terms};
\end{tikzpicture}
```

See

$$\underbrace{a \qquad b \qquad c}_{\text{three terms}}$$

Reverse the path or use `mirror` to change the side on which the brace opens. Keep the label outside the grouped objects.

13.8 Decorating Curved Paths

The base path need not be straight.

```
\draw[
  decorate,
  decoration={snake,amplitude=1mm}
] (A) to[out=35,in=145] (B);
```

TikZ first determines the curved geometry and then applies the decoration along it. If the result is too busy, simplify either the curve or the decoration.

13.9 Worked Project: Three Explanatory Roles

The project uses one brace, one spring, and one boundary. Each has a different meaning.

13.9.1 Stage 1: Group Three Quantities

```
\node at (0,0) {$a$};
\node at (1,0) {$b$};
\node at (2,0) {$c$};
\draw[decorate,decoration={brace,mirror}]
  (-.3,-.4) -- (2.3,-.4);
```

13.9.2 Stage 2: Add a Flexible Connection

```
\draw[decorate,decoration={
  coil,segment length=5mm,amplitude=2mm
}] (3,0) -- (6,0);
```

13.9.3 Stage 3: Add a Schematic Boundary

```
\draw[decorate,decoration={
  zigzag,segment length=5mm,amplitude=1.5mm
}] (0,-2) -- (6,-2);
```

13.9.4 The Complete Picture

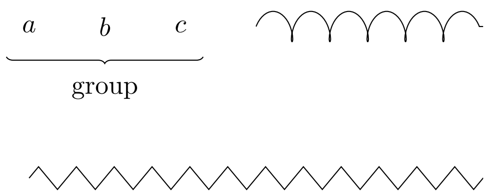
```
\begin{tikzpicture}
  \node at (0,0) {$a$};
  \node at (1,0) {$b$};
  \node at (2,0) {$c$};

  \draw[decorate,decoration={brace,mirror}]
    (-.3,-.4) -- (2.3,-.4)
    node[midway,below=6pt] {group};

  \draw[decorate,decoration={
    coil,segment length=5mm,amplitude=2mm
  }] (3,0) -- (6,0);

  \draw[decorate,decoration={
    zigzag,segment length=5mm,amplitude=1.5mm
  }] (0,-2) -- (6,-2);
\end{tikzpicture}
```

See



Design

The ordinary geometry remains simple. Decoration choices distinguish grouping, flexibility, and boundary information.

13.10 Debugging Workshop

13.10.1 Forgetting `decorate`

Incorrect:

```
\draw[decoration={coil}] (0,0) -- (5,0);
```

Correct:

```
\draw[decorate,decoration={coil}] (0,0) -- (5,0);
```

13.10.2 Loading Only One Decoration Library

Path-morphing decorations and braces come from different libraries. Load both when a figure uses both families.

13.10.3 Using Excessive Amplitude

Begin with a small value appropriate for print. Increase it only when the decoration is difficult to recognize.

13.10.4 Decorating Every Path

Reserve decorations for paths whose roles differ from ordinary connections.

13.11 Exercises

1. Draw the same segment as a coil and as a zigzag.
2. Compare two coil amplitudes with identical endpoints.
3. Compare two zigzag segment lengths with identical amplitudes.
4. Draw a snake along a curved path.
5. Group four mathematical objects with a labeled brace.
6. Move a brace to the opposite side with `mirror`.
7. Define a reusable style for a schematic boundary.
8. Remove one unnecessary decoration from the worked project and explain why the ordinary line is sufficient.

13.12 Solutions

13.12.1 Exercises 1–4

```
\draw[decorate,decoration={coil}] (0,0) -- (5,0);
\draw[decorate,decoration={zigzag}] (0,-1) -- (5,-1);
```

```
\draw[decorate,decoration={coil,amplitude=1mm}]
  (0,0) -- (5,0);
\draw[decorate,decoration={coil,amplitude=2mm}]
  (0,-1) -- (5,-1);
```

```
\draw[decorate,decoration={zigzag,segment length=3mm}]
  (0,0) -- (5,0);
\draw[decorate,decoration={zigzag,segment length=7mm}]
  (0,-1) -- (5,-1);
```

```
\draw[decorate,decoration={snake}]
  (A) to[out=35,in=145] (B);
```

13.12.2 Exercises 5–7

```
\draw[decorate,decoration={brace,mirror}]
  (0,0) -- (4,0)
  node[midway,below=6pt] {four objects};
```

Exercise 6 adds or removes `mirror`; keep every coordinate unchanged.

```
\tikzset{
  boundary/.style={
    decorate,
    decoration={zigzag,amplitude=1mm}
  }
}
```

For Exercise 8, replace one decorated path with an ordinary `--` path; do not change its endpoints.

13.13 ChatGPT Workshop

Ask ChatGPT to preserve the underlying path geometry while evaluating the decorations.

1. Explain separately what `decorate`, the decoration name, `segment length`, and `amplitude` do in this path.
2. Compare coil, zigzag, snake, and bumps for this mathematical role and recommend the least distracting choice.
3. Reduce an excessive decoration without changing the path endpoints.
4. Diagnose why this brace does not appear and check both the library and `decorate` option.
5. Identify decorations that are merely ornamental and replace them with ordinary paths.

13.14 Final Checklist

Before continuing to Chapter 14, check that you can

1. load both decoration libraries used in this chapter;
2. activate and select a path decoration;
3. distinguish coils, zigzags, snakes, and bumps;
4. control repetition length and amplitude;
5. draw and label a brace;
6. move a brace with `mirror`;
7. decorate a curved path; and
8. preserve simple geometry beneath a meaningful effect.